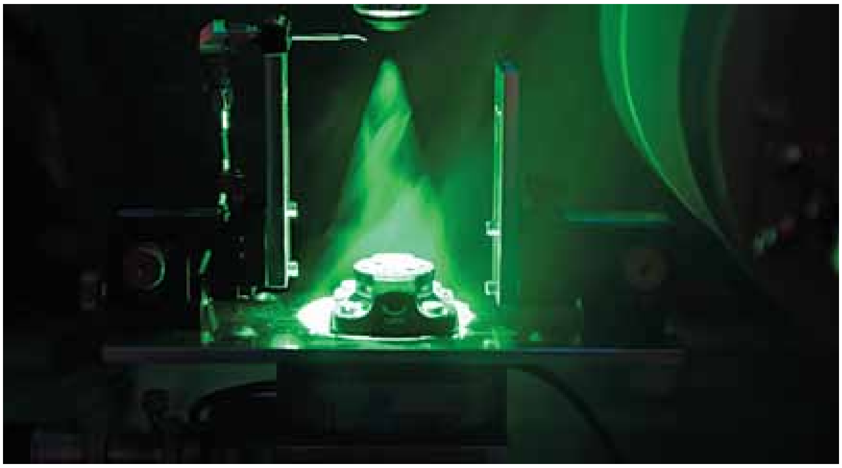
Nanoscale device for spin technology

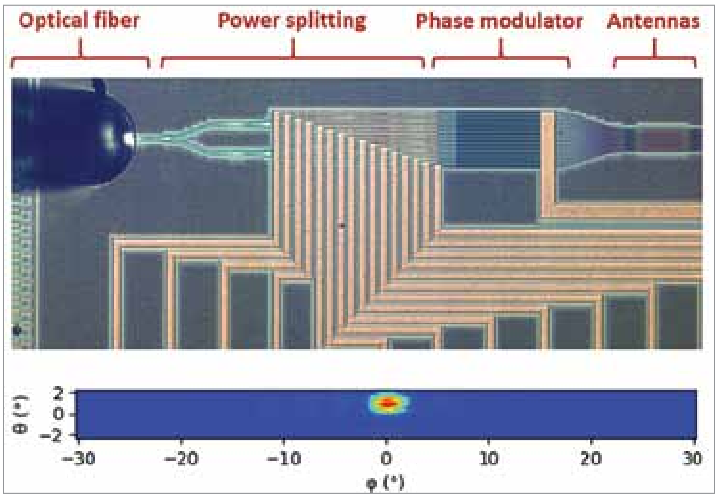


Magneto-optical microscope used for imaging spin waves in a Fabry-Pérot resonator (Credit: www.aalto.fi/en)

Traditional electronic devices use electrical charge to carry out power-intensive computations. But since more heat is created as more calculations are performed faster, there is a concern relating to the limits of how small and fast chips can get before overheating. This is where spintronics comes in. Researchers at Aalto University have developed a new device for spintronics, marking a step towards the goal of using spintronics to make computer chips, and data processing and communication devices small and powerful. The device is a Fabry-Pérot resonator that creates controlled light beams, which control and filter waves of spin in devices that are only a few hundreds of nanometres across.

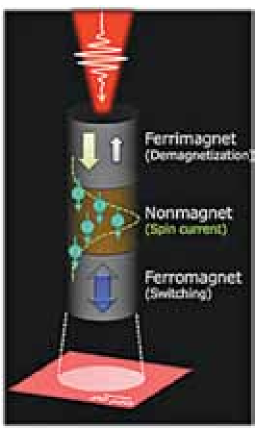
Algorithm calibration towards developing advanced LiDAR systems

CEA-Leti has developed genetic algorithms to calibrate high-channel-count optical phased arrays (OPAs) as well as an advanced measurement setup that enables wafer-scale OPA characterisation. OPAs are an emerging technology made of arrays of closely spaced (around 1µm) optical antennas, which radiate coherent light in a broad angular range. The produced interference pattern can then be modified by adjusting the relative phase of the light emitted by each antenna, enabling solid-state beam steering. This can improve performance in scanning speed, power efficiency, and resolution compared to the heavy, power-hungry, and expensive mechanical beam-steering systems used in current LiDARs.



Microscope view of a silicon photonics based optical phased array. Far field emission profile of the OPA for a calibration at 0° (Credit: www.leti-cea.com)

Ultra-fast technology for energy-efficient data storage



A schematic illustration of the demonstrated ultrafast and energy efficient switching of ferromagnet driven by a single femtosecond laser pulse (Credit: www.tohoku.ac.jp/en)

The amount of data being generated today is continuously growing. If existing technologies remained constant, then all the global electricity consumption would be devoted entirely to data storage by 2040. To drastically reduce energy consumption required for data storage, researchers at the Université de Lorraine in France and Tohoku University have developed an innovative technology that utilises an ultra-fast laser pulse whose duration is as short as 30 femto seconds or 0.00000000000003 seconds. This was driven by a single laser pulse without a switching of the ferromagnetic layer.